Envenomation

Support Adequate Airway, Breathing
Using appropriate CPG

Manage signs of shock
Treat as per non-traumatic shock CPG

Go to Non-Traumatic Shock CPG

If signs of allergic reaction/ anaphylaxis
Treat as per allergy CPG

Go to Allergy CPG

General Care

- Remove constricting items (e.g. rings) from area of bite/swelling.
- Mark area of swelling and record progression over time.
- Irrigate with sterile saline or clean water

Insect Bite

- Remove stinger if visible.
- Use ice for symptomatic relief.

EMT

Snake Bite

- Immobilize and splint the injured extremity.
- Do not apply ice to snake bites.
- Do not incise wound or try to "suck" the venom out.

EMT

Marine Animal Envenomation

- Remove any pieces of debris or stingers with tweezers or gloved hand.
- If the stinger or tentacle is not able to be removed easily with gentle traction, do not compress with bandages as additional envenomation may occur.
- Vinegar (if available) may be used for Jelly-fish stings.
- Rinse a small portion of the affected area with hot water. If the patient gets relief, continue with a larger area and then progressively to the entire area.
- Bandage loosely and splint injured extremity

EMT

Reassess

Patient Care Goals

1. Rapidly identify and manage suspected envenomation to prevent systemic complications and facilitate time-sensitive interventions.
2. Minimize secondary injury through supportive care, including airway management, pain control, and prevention of venom spread.
3. Ensure rapid transport to an appropriate medical facility for definitive evaluation and treatment, such as antivenom administration when indicated.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients presenting with signs and symptoms suggestive of envenomation from insect bites, snake bites, or marine animal stings, including:
  - Localized pain, swelling, redness, or bruising at the site of injury.
  - Systemic symptoms such as nausea, vomiting, muscle cramps, weakness, respiratory distress, or neurological changes (e.g., paralysis, convulsions).
  - History of exposure to venomous animals (e.g., snakes, spiders, scorpions, jellyfish, or stingrays).

Exclusion Criteria

- Bites or traumatic injuries from non-venomous animals such as dogs or cats.

Key Patient Assessment Points

1. **Envenomation Screening**
   - Obtain a detailed history: Type of animal/insect involved (if known), time of exposure, location of bite/sting, and progression of symptoms.
2. **Perform a Focused Assessment**
   - Inspect the site: Look for puncture wounds, fang marks, stingers, tentacles, or embedded spines.
   - Vital signs: Monitor for tachycardia, hypotension, respiratory rate changes, or fever.
   - Neurological: Assess for weakness, paralysis, fasciculations, altered mental status, or seizures.
   - Respiratory: Evaluate for bronchospasm, stridor, or hypoxia.
   - Cardiovascular: Check for shock, arrhythmias, or edema.
3. **Consider Alternative Causes**
   - Consider differentials: infections (e.g., cellulitis), trauma.
4. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
     - Red Priority: Severe: Systemic toxicity (e.g., neurotoxicity, hemotoxicity, respiratory failure, anaphylaxis).
     - Yellow Priority: Moderate: Progressive swelling, mild systemic effects (e.g., nausea, mild weakness).
     - Green Priority**:** Mild: Localized pain/swelling without systemic symptoms.

Patient Management & Interventions

1. **Scene Safety and Patient Removal**
   - Ensure scene safety; do not engage in search for the animal. Remove patient to a safe environment (away from animals or water) before treatment.
2. **Primary & Secondary Survey**
   - Perform Primary and Secondary Surveys as per Universal Care CPG.
3. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
4. **Supportive Care**
   - Maintain adequate Airway, Breathing and Circulation as per Universal Care CPG.
   - Treat anaphylaxis as per Allergy / Anaphylaxis CPG.
5. **General Management**
   - Remove constricting items (e.g. rings) from area of bite/swelling.
   - Mark area of swelling and record progression over time.

- Irrigate with sterile saline or clean water

6. **If: Insect Bite/Sting**
   - Remove stinger if visible (scrape with edge, do not squeeze).
   - Apply ice for symptomatic relief.
   - Monitor for anaphylaxis – Treat as per Allergy / Anaphylaxis CPG if required.
   - Scorpion & Spider Bites:
     - Monitor for autonomic storm (hypertension, salivation, arrhythmias)
     - Benzodiazepines may be required for agitation.
7. **If: Snake Bite**
   - Immobilize and splint the injured extremity.
   - Use pressure bandage technique for possible neurotoxic snakes.
   - Do NOT apply ice to snake bites.
   - Do NOT incise wound or try to "suck" the venom out.
   - Do NOT wash or clean the bite site:
     - Venom residues on the skin or fang marks are often needed for snake venom detection kits (used in labs/hospitals to identify the snake species and guide antivenom choice).
     - Washing may remove venom traces and make identification difficult.
     - Exception: If there's gross contamination with dirt, mud, or foreign material → gentle cleaning around (not scrubbing the fang punctures themselves).
8. **If: Marine Animal Envenomation**
   - Remove any pieces of debris or tentacles (gloved hands or forceps).
     - If the stinger or tentacle cannot be removed easily with gentle traction, do not compress with bandages as additional envenomation may occur.
   - For jellyfish/stingers: Rinse with vinegar or seawater (not freshwater).
   - Use hot water immersion (40-45°C) for 20-30 minutes if tolerated.
     - Rinse a small portion of the affected area with hot water. If the patient gets relief, continue with a larger area and then progressively to the entire area.
   - Bandage loosely and splint injured extremity
9. **Pain Management**
   - Administer analgesics as per Pain Management CPG.
10. **Patient Reassessment**
    - Frequently Reassess the patient as per Universal Care CPG.
    - Specific Patient Reassessment Points:
      - Reassess redness/swelling every 15 minutes.

Patient Safety Considerations

- Time-Sensitive Transport
  - Patients with suspected severe envenomation or worsening symptoms should be rapidly transported to a facility capable of antivenom administration.
  - Clearly communicate with Dispatch identifying envenomation type, symptoms, and onset time.
- Avoid Harmful Interventions
  - Prohibit traditional remedies (e.g., cutting, sucking venom, tourniquets) as they can worsen outcomes.

Clinical Insights & Additional Guidance

- Snake bites
  - Most snakebites are "dry," i.e., no venom is injected.
  - If envenomated, some of the following should occur in 5–30 minutes:
    - Severe burning pain out of proportion to the wound.
    - Edema around the bite out of proportion to the wound.
    - Small, non-blanching purple spots (petechiae), bruising, or continued oozing from site.
    - Numbness or tingling of the mouth, extremities, or bite site.

        - Metallic taste in the mouth.
        - Involuntary twitching of the mouth, extremities, or bite site.
        - Generalized Weakness
    - Exotic snakes (Cobra, Krait, etc.) or Coral may cause neurologic and respiratory depression prior to a local reaction. Observe for mental status change, respiratory depression, convulsions, or paralysis.
- Marine Animal Envenomation
    - Portuguese man-of-war, although often mistaken for a “Jellyfish”, is treated differently than most Coelenterates, using hot water and not vinegar.
    - Stonefish envenomation can cause systemic toxicity with hypotension, tachycardia, cardiac arrhythmias, diaphoresis, dyspnea and pulmonary edema.
    - Most cases are successfully managed with hot water immersion and symptomatic care, however some may require a specific antivenom.

Documentation

- Identification: Patient’s full name, age, and gender.
- Mechanism / Medical Complaint: Description of bite/sting, animal type (if known), time of exposure.
- Injuries / Information: Site of injury, symptoms progression, envenomation mimics considered (e.g., allergy, infection).
- Signs: Wound appearance, swelling measurement, vital signs, neurological exam, respiratory status.
- Treatment:
    - Interventions (e.g., immobilization, hot water, ice, oxygen).
    - IV access and fluids administered.
    - Pain management provided.
    - Hospital notification and envenomation alert.
- Allergies: Known allergies to antivenom or insects.
- Medications: Document any prior treatments or chronic meds.
- Background: Relevant history (e.g., previous envenomations, comorbidities like coagulopathies).
- Other:
    - Scene details and witness statements.
    - Accurate timeline of exposure and symptom onset.
    - Communication with poison control or receiving facility.

Key Performance Indicators

1. Assess and document bite type and severity.

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